

Serial ATA: Hit or Miss?

With the upcoming ATA/133 standard pushing the limits of the maximum transfer rates supported by the Parallel ATA interface, the next step to revolutionise hard disk interfaces will be Serial ATA. It offers greater speed, easier fabrication and application.

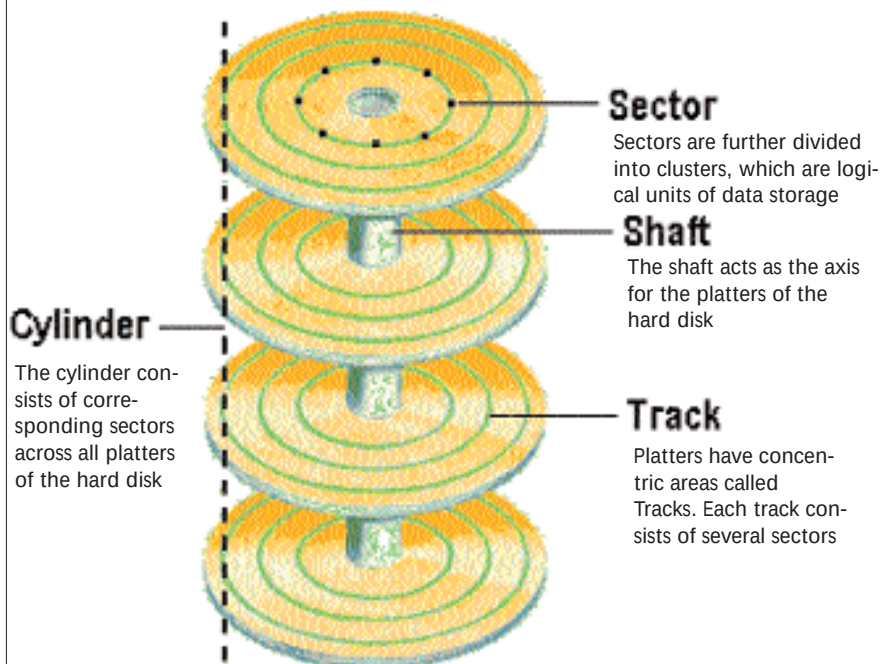
The Serial ATA interface is to ATA what FireWire is to SCSI—the entire approach is to convert the parallel standard that is used by existing ATA interfaces to a serial format (where one bit is transferred at a time) so that higher data transfer frequencies can be supported. Also, this approach greatly simplifies the fabrication of cables and connecting wires that are used by the interface, since there are fewer signals to deal with, facilitating a less cluttered cabinet.

This translates into easier airflow within the system—and, subsequently,

more efficient cooling. In fact, Serial ATA uses just four signal pins (there are a total of seven pins; the others are used for transmitting overhead commands) that actually carry the data and include the power and ground pins. This is significantly lower than the 80 lines that are used in today's ATA/100 and ATA/133 cables.

This standard will support transfer rates of up to 150 MBps. Also, since this is a point-to-point protocol, there will be no place for master and slave devices on a single cable—and bandwidth will not need to be shared between devices. Another big advantage with this standard is that it is backward-compatible with the existing parallel ATA standard at the driver level. Therefore, all existing applications that use the parallel ATA standard can be fully extended to support Serial ATA.

Under the bonnet of a Hard Disk



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